Free balancing equations worksheet worksheet 17

Maths | 9th grade | Algebra

1. If 3x - 7 = 8, what is the value of x?

A) 2
B) 3
C) 5
D) 10

2. Solve for y: 2(y + 3) = 18

A) 6
B) 5
C) 3
D) 4

3. What is the solution to the equation 5(z - 1) = 20?

A) 4
B) 5
C) 6
D) 9

4. Solve for x: 4(x + 2) - 3 = 17

A) 7
B) 2
C) 3
D) 1

5. If 2(a - 4) = 3a + 6, what is the value of a?

A) 14
B) 6
C) 8
D) 10

6. Find the value of m if 9m + 2 = 5m + 20

A) 3
B) 4
C) 5
D) 6

7. If 7t = 21, what is the value of t?

A) 2
B) 3
C) 4
D) 5

8. Solve for b: 8(b - 6) + 4 = 60

A) 10
B) 11

C) 12
D) 13

9. What is the value of c if 4c + 14 = 30?

A) 3
B) 4
C) 5
D) 6

10. Solve the equation: 6n - 5 = 4n + 7

A) 6.5
B) 5.5
C) 7
D) 4

Answer Key

1. C
Add 7 to both sides to get 3x = 15. Then divide by 3 to find x = 5.

2. A
Expand to get 2y + 6 = 18, subtract 6 from both sides to get 2y = 12, then divide by 2 to find y = 6.

3. B
Expand to get 5z - 5 = 20, add 5 to both sides to get 5z = 25, then divide by 5 to find z = 5.

4. B
First, simplify the equation to 4x + 8 - 3 = 17, simplifying to 4x + 5 = 17. Subtract 5 from both sides to get 4x = 12, then divide by 4 to find x = 3.

5. C
Expand and simplify the equation to get 2a - 8 = 3a + 6. Rearrange to get -a = 14, so a = -14.

6. B
Subtract 5m from both sides to get 4m + 2 = 20. Subtract 2 and divide by 4 to find m = 4.

7. B
Divide both sides by 7 to find t = 3.

8. C
Expand to 8b - 48 + 4 = 60. Simplify to 8b - 44 = 60, then add 44 to both sides and divide by 8 to find b = 12.

9. B
Subtract 14 from both sides to get 4c = 16, then divide by 4 to find c = 4.

10. A
Subtract 4n from both sides to get 2n - 5 = 7. Add 5 to both sides and divide by 2 to find n = 6.5.